

Evidence-Constrained Diagnostic Assistance for Unlabeled Neutron-Current Time Series: A Tool-Using Workflow and Retrieval Robustness Study

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Abstract

Neutron-instrumentation records from nuclear power plants are high-volume raw time series that commonly lack a uniform sample-level fault ground truth, whereas their interpretation must remain traceable to authoritative technical evidence. We present an evidence-constrained diagnostic-assistance workflow rather than an autonomous fault classifier. A zero-prior, tool-using agent first converts unlabeled neutron-current CSV records into an auditable condition report by applying deterministic probes for data health, morphology, anomalies, lag, extremes, and spatial snapshots. A compact structured alert summary then drives multi-stage retrieval over nuclear instrumentation and control manuals, enabling a citation-bearing diagnostic memo. We evaluate the retrieval layer under controlled expansion of a fixed IAEA core library with same-domain supplementary documents, using Chinese and English queries, chunking configurations, and both strict chunk and document-level relevance criteria. In the evaluated setup, cosine similarity can increase while the priority of pre-designated IAEA sources decreases; short Chinese queries are especially sensitive to corpus expansion, whereas the tested chunk-size and overlap changes have a smaller effect. The results show that similarity alone is insufficient to govern evidence retrieval in this setting and motivate terminology-rich queries, explicit source-priority policies, and human review of diagnostic conclusions.

Keywords: neutron instrumentation, online monitoring, diagnostic assistance, retrieval-augmented generation, evidence traceability, retrieval robustness

1. Introduction

Nuclear power plants continuously acquire neutron and instrument-channel measurements for surveillance and condition monitoring. These raw time series can reveal changes in detector behavior, signal integrity, and operating regimes, but the CSV records used here do not provide a uniform sample-level fault ground truth. Technical interpretation therefore continues to rely on authoritative guidance for online monitoring (OLM) and instrument-channel assessment^[1,2,3]. Neutron-noise analysis further provides a non-intrusive route to studying reactor and instrumentation phenomena^[4,5]. Although data-driven diagnostic methods can be effective for predefined scenarios, their assumptions of fixed feature schemas and labeled fault classes are not always met by plant archives^[6,7].

Large language models (LLMs) can help organize heterogeneous observations and technical knowledge, but free-form generation is unsuitable as a decision authority in a safety-relevant setting. Retrieval-augmented generation (RAG) conditions an answer on external evidence^[8]; in nuclear applications, its value depends on whether the retrieved manuals genuinely support the statement being made. Recent nuclear RAG studies emphasize factuality,

provenance, and failure modes such as incorrect retrieval and unsupported interpretation^[9,10,11,12]. A practical open question is how to connect measurement-side findings to that evidence layer without asking an LLM to infer a root cause directly from a raw CSV file.

This work addresses that question through an evidence-constrained diagnostic-assistance workflow. A tool-using agent observes unlabeled raw neutron-current series with minimal diagnostic priors and writes an auditable condition report. This report is a derived analysis artifact, not a ground-truth annotation. It is then compressed into a structured alert summary that drives manual retrieval and citation-bearing explanation. The intended output is not an automatically confirmed fault diagnosis: it distinguishes observed data phenomena from manual-supported candidate interpretations, both of which remain subject to engineering review.

The reliability of this workflow depends critically on retrieval. A manual library may need to include both a small, pre-designated authoritative set and a larger set of same-domain supplements. Adding relevant supplements can improve apparent nearest-neighbor similarity while reducing the priority of the authoritative sources specified by a deployment policy. This tension is especially important for bilingual technical terminology and for overlap-based chunking, where a useful passage may not have the identical

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Evidence-constrained diagnostic assistance

A report-centered architecture that separates observed signal events, documentary evidence, and engineering judgment.

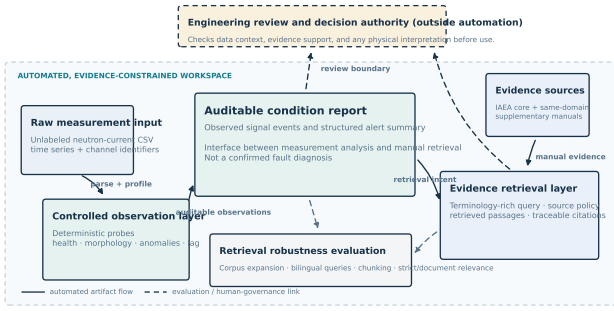


Figure 1: Evidence-constrained diagnostic-assistance architecture. The condition report and structured alert summary form the auditable interface between raw measurement analysis and manual retrieval. The evaluation module stress-tests the retrieval layer under corpus, query, and chunking variation; engineering review remains outside the automated chain and retains decision authority.

chunk identifier used as a query probe.

The contributions are threefold. First, we develop a zero-prior, tool-using workflow with a persistent workspace contract for converting neutron-current CSV data into auditable observations. Second, we introduce report-driven multi-stage RAG that maps structured observations to citable manual evidence. Third, we evaluate evidence retrieval under controlled core-supplement expansion, query language, chunking, and strict versus document-level relevance. The resulting evidence is used to characterize deployment boundaries rather than to claim autonomous fault identification. The complete transformation contract is specified in section 2.

2. System Design and Methodology

This section specifies the evidence-constrained transformation contract used throughout the study: a raw measurement record is converted into an auditable observation packet, compressed into a structured alert summary, linked to retrieved manual evidence, and expressed as a citation-bearing memo for engineering review. The automated chain produces derived observations and candidate interpretations; engineering review remains outside that chain and retains decision authority.

2.1. Data Resources

The system consumes two complementary data resources: a plant neutron-current measurement archive that supplies observable signal phenomena, and a technical evidence corpus that supplies retrievable documentary support. Their roles are deliberately distinct: neither resource alone is treated as a fault-label database.

2.1.1. Neutron-current measurement archive

The measurement archive contains 42 multi-channel neutron-current CSV records from four instrumentation-bank groups (A1, A2, B1, and B2) in the same plant campaign. Each record contains one time column and seven neutron-current channels (eight columns overall). The archive composition is summarized in table 1.

Table 1: Composition of the neutron-current measurement archive.

Group	Number of series	Channels per series	Columns per series
A1	11	7	8
A2	11	7	8
B1	10	7	8
B2	10	7	8
Total	42	—	—

Channel identifiers follow plant tagging conventions (e.g., RIC...MA_VER). Within a group, the seven detectors occupy a circumferential layout, so their ordering in the CSV is associated with relative channel position rather than an arbitrary list. Representative series contain on the order of 6×10^5 samples at a nominal 1 s interval over multi-day windows.

The raw exports can contain local timestamp disorders and non-monotonic segments. Chronological sorting is therefore a mandatory first operation. The present workflow does not use a uniform sample-level fault ground truth: its output is a condition report of observed signal events, not a supervised fault class. Derived reports are workflow artifacts rather than annotations attached to the raw CSV records.

2.1.2. Technical evidence corpus

The evidence resource is a curated 64-document collection of nuclear instrumentation and control manuals converted from PDF to Markdown and accompanied by document-source metadata. It contains a 13-document IAEA core, including guidance on online monitoring and instrument-channel assessment^[1,2], together with 51 same-domain supplementary documents from sources such as NRC, EPRI/DOE, OECD/NEA, and laboratory or topical reports. The historical and CPU protocols use distinct index and chunk constructions, as detailed in section 3, but draw from this same document collection.

The supplementary documents are not assumed to be irrelevant. They enlarge the searchable knowledge base, while the IAEA core represents a deployment policy that prioritizes a pre-designated authority set. This distinction motivates the source-priority measurements in the evaluation but does not redefine source priority as universal relevance.

2.1.3. Data roles and semantic boundary

The two resources meet only through an auditable semantic interface. Deterministic probes first convert a measurement record $\mathbf{X}$ into observed events and a structured

alert summary $\mathbf{a} = g(\mathbf{X})$. Retrieval then converts $\mathbf{a}$ into terminology-rich queries over the evidence corpus. Thus a measurement anomaly is not directly promoted to a physical diagnosis, and a manual passage is not treated as a direct explanation of an unprocessed waveform. The resulting evidence-linked memo remains an aid to engineering review.

2.2. Measurement-side observation construction

2.2.1. Objective and zero-prior scope

The measurement-side objective is to convert one raw neutron-current CSV record into a structured, human-readable condition report. “Zero-prior” does not mean that the system ignores the data format or the relative channel layout described in section 2.1. Rather, it does not assume a labeled fault library, a pre-specified mapping from an observed pattern to a physical cause, or handcrafted diagnostic frequency bands. The workflow treats channel identifiers as operational identifiers and uses deterministic statistics to expose morphology, redundancy, anomalies, lag, extremes, and cross-channel snapshots. Its output is an observation record, not a confirmed fault class.

2.2.2. Workspace initialization and data health

The load-and-profile probe is the mandatory entry point. Given a CSV path, it creates a dedicated workspace, parses and chronologically sorts the time column, writes a cleaned active record, and records shape, nominal sampling interval, missingness, and inactive-channel indicators in a persistent blackboard (`state.json`). All later probes resolve the active record from this workspace rather than accepting a new free-text data path. This establishes one analysis unit per record and preserves the provenance of every subsequent observation.

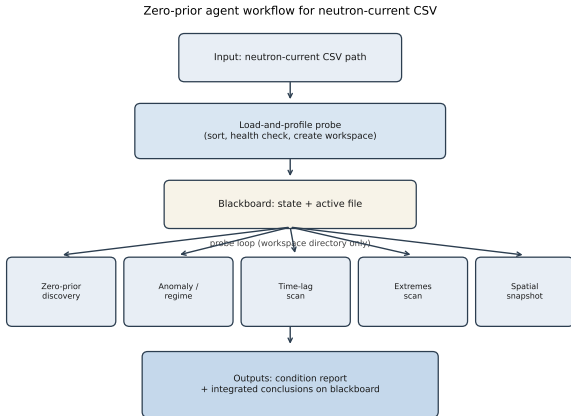


Figure 2: Measurement-side observation workflow. Profiling initializes a workspace blackboard; deterministic probes operate on the active record and accumulate an auditable condition report.

2.2.3. Deterministic observation probes

After initialization, the planner may select only the registered probes in table 2. The discovery probe performs

a global longitudinal scan: uniqueness and coefficient of variation assign coarse morphological roles, while pairwise correlations identify near-redundant pairs and strongly co-moving groups. Given known relative channel placement, such co-movement can motivate inspection of spatial proximity or shared instrumentation context; it is not, by itself, a physical diagnosis.

The anomaly/regime probe segments channels into steady and transient regimes and enumerates spike episodes and isolated zero events. The time-lag probe distinguishes lead-lag patterns from near-synchronous groups. Extremes are localized with their timestamps, and the snapshot probe records a multi-channel cross-section at a selected time. Together, these operations form a reproducible inventory of signal events and relationships.

Table 2: Deterministic probes used to construct the condition report. The implementation identifier is included for reproducibility.

Probe	Observation role	Implementation ID
Load-and-profile	Initializes the workspace, sorts timestamps, and records data-health indicators.	<code>load_and_profile_csv_tool</code>
Zero-prior discovery	Identifies morphology and co-moving or near-redundant channel groups.	<code>zero_prior_discovery_tool</code>
Anomaly and regime	Segments regimes and enumerates spikes and isolated zero events.	<code>anomaly_detector</code>
Time-lag scan	Estimates lead-lag versus near-synchronous channel groups.	<code>global_time_lag_scan_tool</code>
Extremes scan	Localizes global maxima/minima and their timestamps.	<code>global_extremes_scan_tool</code>
Spatial snapshot	Captures the multi-channel state at a selected time.	<code>get_spatial_snapshot_tool</code>

2.2.4. Operational definitions and auditable interface

Let $\mathbf{X} = \{x_{t,c}\}$ denote a chronologically sorted multi-channel record, where t indexes the sample and c indexes the detector channel. The system does not ask the planner to estimate these quantities. Deterministic probes compute them and write their results to the workspace blackboard; the LLM only selects registered probes and organizes the resulting evidence.

For each channel, the anomaly probe uses a centered rolling window W_t of 31 samples. Its local baseline and robust local deviation are

$$m_{t,c} = \text{median}_{\tau \in W_t} x_{\tau,c}, \quad (1)$$

$$d_{t,c} = \text{median}_{\tau \in W_t} |x_{\tau,c} - m_{\tau,c}|.$$

The implementation marks a sample as a spike candidate only when both a local and a global-amplitude condition hold:

$$I_{t,c}^{\text{spike}} = \mathbb{I}(|x_{t,c} - m_{t,c}| > 5d_{t,c}) \mathbb{I}(|x_{t,c} - m_{t,c}| > 0.05R_c),$$

$$R_c = \max_t x_{t,c} - \min_t x_{t,c}. \quad (2)$$

Contiguous positive samples are merged into one spike episode. An isolated zero event is separately recorded by $I_{t,c}^{\text{zero}} = \mathbb{I}(x_{t,c} = 0)$. These rules define detected signal events, not physical fault labels.

For a channel pair (i, j) , the discovery probe computes Pearson correlation

$$r_{ij} = \frac{\sum_t (x_{t,i} - \bar{x}_i)(x_{t,j} - \bar{x}_j)}{\sqrt{\sum_t (x_{t,i} - \bar{x}_i)^2} \sqrt{\sum_t (x_{t,j} - \bar{x}_j)^2}}. \quad (3)$$

The current implementation flags near-redundancy at $r_{ij} \geq 0.99$ and strong co-movement for $0.85 \leq r_{ij} < 0.99$. Together with lag, extremes, and spatial-snapshot outputs, these quantities form a structured alert interface

$$\mathbf{a} = g(\mathbf{X}) = \{\mathbf{a}_{\text{health}}, \mathbf{a}_{\text{morph}}, \mathbf{a}_{\text{anom}}, \mathbf{a}_{\text{lag}}, \mathbf{a}_{\text{spatial}}\}, \quad (4)$$

which is passed to retrieval. This separation prevents free-form text from becoming the sole interface between measurement analysis and manual-grounded explanation.

2.2.5. Observation packet, condition report, and structured alert summary

Each completed probe contributes to an observation packet that records the analysis scope, the registered probe and its operational parameters, observed signal events, and relevant uncertainty or data-health cues. The human-readable condition report preserves these results in chronological form for audit and engineering inspection. Its compact alert summary retains only the fields needed downstream—for example, data-health indicators, morphology, spike and zero summaries, co-movement, lag, extremes, and snapshot cues. This report/summary separation preserves auditability without forcing retrieval to operate on a long, unstructured narrative; neither artifact is a fault label or a confirmed diagnosis.

2.3. Report-driven evidence retrieval

The condition report serves two audiences. Its chronological narrative and optional JSON snapshots support human inspection and audit, whereas its compact alert summary is the machine-facing interface to retrieval. The summary contains variable-wise alerts, spike and zero-event counts, morphology, collinearity, lag, and selected spatial-snapshot cues. Queries are constructed from this structured representation rather than from a short operator phrase or an unreviewed free-text diagnosis hypothesis.

Source PDFs are page-cropped when required, converted to Markdown, chunked, and embedded in a vector store^[13]. At retrieval time, the system first searches for passages relevant to the observed phenomenon or mechanism, then may supplement them with passages on redundancy, topology, monitoring practice, or recommended checks. A memo composer links candidate interpretations to the retrieved passages with numbered citations. It must not convert this support into an automatically confirmed root cause.

This design makes the report the semantic bridge between the measurement and evidence resources introduced in section 2.1. It also explains why the retrieval experiments in section 3 are part of the system contribution: they stress-test whether this evidence layer preserves the intended source policy under controlled corpus expansion, query variation, and chunking.

2.4. Constrained tool use and runtime reliability

The workflow treats runtime reliability as a methodological requirement rather than an implementation afterthought. In early multi-step prototypes, the planner was allowed to restate absolute paths, emit unrestricted tool calls, and pass lengthy condition text directly into downstream diagnosis. This produced path drift, malformed structured output, invented analyzers, and poorly grounded diagnostic prose. The final protocol converts these observed failure modes into explicit runtime constraints, summarized in table 3.

Table 3: Observed failure modes in early tool-using prototypes and runtime constraints in the final protocol.

Failure mode	Runtime constraint
Path restatement across tool steps	Only load-and-profile accepts a CSV path; later probes receive a workspace directory and resolve the active record from the blackboard.
Markdown-wrapped, truncated, or malformed tool JSON	Strip Markdown fences; extract the first brace-balanced JSON object or array; validate the requested action before execution.
Invented or unregistered analyzer names	Accept only registered probe identifiers; unknown actions fail closed.
Oversized condition narratives for downstream reasoning	Retain a human-readable report for audit but expose a compact structured alert summary to retrieval.
Ungrounded diagnostic prose	Construct retrieval from the alert summary and require evidence-linked, citation-bearing memo output.

2.4.1. State ownership and constrained actions

The local LLM planner selects the next analysis action, but it does not own the authoritative state. The Python router executes registered analyzers and updates the workspace blackboard. Once initialization has completed, probes cannot silently switch to another raw or sorted file because the active record is resolved from that state. Missing initialization causes the workflow to fail closed. This moves data-path ownership from mutable model strings to a runtime contract.

At the action interface, the router tolerates common presentation noise while retaining a narrow execution boundary. It extracts the first brace-balanced JSON object or array even when the model adds Markdown fences or short commentary, then validates that the `action` field names a registered probe. The protocol does not claim perfect structured generation; instead, it makes malformed or invented actions visible and non-executable^[14,15,16].

2.4.2. Evidence-constrained output boundary

The runtime constraints continue beyond tool execution. A multi-page condition narrative is useful for audit but unsuitable as the sole retrieval query, because it mixes many observations with incidental detail. The system therefore exposes a compact alert summary to the retrieval layer and requires memo claims to remain linked to retrieved passages. This reduces the opportunity for fluent but unsupported interpretation^[17]. The automated chain produces observations and evidence-linked candidate interpretations; an engineer remains responsible for judging context, support adequacy, and any physical conclusion before use.

3. Evaluation Protocol

3.1. Evaluation questions and scope

The evaluation assesses the reliability of the *evidence-retrieval layer*, not diagnostic accuracy, root-cause identification, or generation-side factuality. It asks whether retrieval maintains a deployment policy that prioritizes a fixed IAEA core when same-domain supplementary manuals are admitted, and how that behavior changes with corpus scope, query formulation, chunking, and encoder choice.

3.2. Corpus, source policy, and protocol variants

The retrieval collection contains 64 technical documents: a fixed 13-document IAEA core and 51 same-domain supplementary documents. The supplementary material is relevant nuclear instrumentation and control evidence, not meaningless noise. The source-priority policy therefore does not assert that IAEA passages are the only relevant passages; it asks whether a deployment that requires a pre-designated authority set can maintain that preference as the searchable corpus expands.

Two protocol variants are reported separately. The historical robustness experiments use `nomic-embed-text` and 17 Markdown index configurations with chunk sizes $\{800, 1000, 1200, 1500\}$ and overlaps $\{50, 80, 100, 120, 150\}$ for valid combinations. The independent CPU encoder benchmark uses 240-word chunks, 24-word overlap, and two MiniLM encoders. Its absolute score scale is therefore not compared directly with the historical index; it tests whether the source-priority phenomenon persists under a separate encoder and chunk construction.

3.3. Queries and controlled corpus expansion

All corpus-scope protocols use the four fixed bilingual queries in table 4. The Chinese entries are represented by Unicode code points solely because the manuscript is compiled with `pdfLaTeX`; the benchmark metadata stores and submits the corresponding UTF-8 strings. The queries intentionally include short core terms and more specific phrase queries, so the experiment evaluates both query language and query formulation rather than claiming an intrinsic language ranking.

Table 4: Fixed queries used in corpus-scope retrieval experiments.

Query type	Chinese code points	English
Core term	U+4E2D U+5B50	neutron
Phrase	U+4E2D U+5B50 U+6D4B U+91CF U+7535 U+6D41	neutron measurement current

Let $\mathcal{D}_{\text{core}}$ be the fixed IAEA pool and let $\mathcal{D}_{\text{sup}}^{(m,t)}$ denote the supplementary documents sampled for expansion size m and trial t . The active retrieval pool is

$$\mathcal{D}_{m,t} = \mathcal{D}_{\text{core}} \cup \mathcal{D}_{\text{sup}}^{(m,t)}, \quad m \in \{0, 5, 10, \dots, 51\}. \quad (5)$$

For $m = 0$, the core-only pool is evaluated once. In the historical index experiments, each non-zero m uses 100 random supplementary-document draws per query and index configuration. In the independent CPU benchmark, each non-zero m uses ten random draws per query and encoder. Retrieval returns the top $k = 10$ chunks.

3.4. Historical robustness and self-retrieval experiments

The historical corpus-scope experiment records top-1, top-3 mean, and top-10 mean cosine scores, IAEA priority@10, and the rank of the first IAEA result. The primary score plotted in fig. 5 is the top-1 cosine score. For a query q and ordered retrieval list $R_k(q) = \{d_1, \dots, d_k\}$,

$$\text{Priority}@k(q) = \frac{1}{k} \sum_{i=1}^k \mathbb{I}(d_i \in \mathcal{D}_{\text{core}}). \quad (6)$$

Priority@k measures adherence to a source-priority policy, not human-judged relevance.

Self-retrieval samples IAEA chunks as probes and forms queries using four modes: `first_sentence`, `title`, `random_window`, and `domain_term`. For the corpus-decay condition visualized in fig. 6, the first-sentence mode is used. The core-only pool uses 80 probes. Each non-zero expansion size uses ten trials of 80 probes each. For a probe chunk n_j^* , its source document $d_j^* = \text{doc}(n_j^*)$, and retrieved set $R_k(q_j)$, strict and document-level recall are

$$\begin{aligned} R_k^{\text{strict}} &= \frac{1}{N} \sum_{j=1}^N \mathbb{I}(n_j^* \in R_k(q_j)), \\ R_k^{\text{doc}} &= \frac{1}{N} \sum_{j=1}^N \mathbb{I}(\exists n \in R_k(q_j) : \text{doc}(n) = d_j^*). \end{aligned} \quad (7)$$

We additionally report mean reciprocal rank, where r_j is the rank of the first relevant result and $r_j = \infty$ when the top- k list has no relevant result:

$$\text{MRR} = \frac{1}{N} \sum_{j=1}^N \begin{cases} 1/r_j, & r_j \leq k, \\ 0, & r_j > k. \end{cases} \quad (8)$$

The strict/document distinction prevents overlap-induced neighboring chunks from being characterized as wholly irrelevant.

3.5. Independent CPU encoder comparison

The independent benchmark compares `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` with `sentence-transformers/all-MiniLM-L6-v2` on CPU. It uses the same 64-document collection, the four queries in table 4, normalized embeddings, top-10 retrieval, 240-word chunks with 24-word overlap, and 11,199 chunks in total. For every trial it reports top-1 and top-10 mean cosine scores, IAEA priority@10, and first-IAEA rank. This benchmark is a direct measured comparison, not an extrapolation from the historical index.

3.6. Aggregation and interpretation boundary

Within each historical index configuration, the stored standard deviations quantify variation across random supplementary-document draws. The error bands in figs. 5 and 6 have a different meaning: each configuration is first reduced to its mean, and the displayed bands are one standard deviation across the 17 available chunking configurations. The CPU comparison reports standard deviations across its ten random document draws. These distinct sources of variation are not pooled.

The experiments establish retrieval behavior for the corpus, encoders, query set, and index grids studied here. They do not establish a universal ranking of languages or chunking strategies, human relevance of every retrieved passage, or the factual correctness of generated memos. Those questions require relevance annotation and generation-side review.

4. Results and discussion

4.1. Traceable case study: from observation to evidence

Across representative workflow executions, the measurement-side probes produce condition reports containing synchronized zero events, spike episodes, and strongly co-moving channel groups. One representative record exhibits pronounced differences in channel morphology, several highly collinear detectors, repeated spike episodes on one channel, and zero dropouts on others. Figure 3 displays downsampled traces from this record; all probe outputs are computed at full resolution.

This result demonstrates that the workflow can produce an auditable observation record and use it to organize manual retrieval. It should not be interpreted as independent confirmation of a physical fault or a root cause. In the present study, the memo is an evidence-linked aid for subsequent engineering review; no uniformly aligned fault-label set is available for estimating end-to-end diagnostic accuracy.

Figure 4 and Table 5 provide a tabulated view of one representative condition-report output. Its seven channels contain 30–34 detected spike episodes (mean 32.4) and each contains 12 isolated zero events. The repeated zero timestamps across channels are a concrete example of why the workflow preserves cross-channel timing rather than

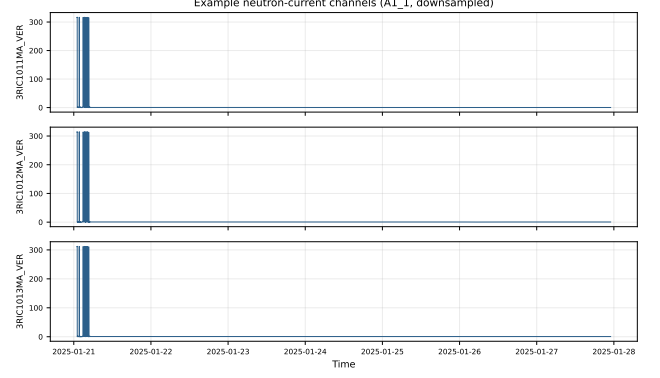


Figure 3: Example neutron-current traces from a representative multichannel record (three channels; downsampled for readability). The full-resolution series is analyzed by the workspace probes.

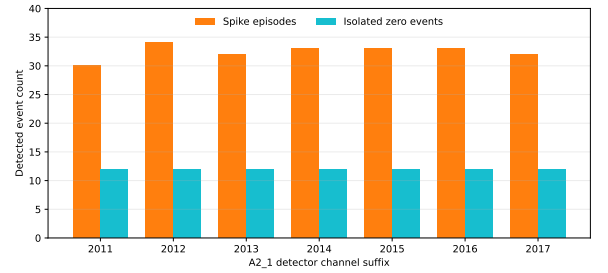


Figure 4: Detected event counts from a representative condition-report output. These are deterministic signal-event counts under eq. (2), not confirmed equipment-fault counts.

converting events immediately into a component-fault label: the observation supports a synchronized signal-quality event, while its physical mechanism requires external review.

4.2. Similarity and source priority can diverge

The corpus-scope results show a consistent deployment risk: increasing the supplementary pool can raise cosine scores while reducing IAEA priority@10. Figure 5 and table 6 aggregate the phrase-query results over all 17 available index configurations. For Chinese, mean IAEA priority@10 falls from 1.00 in the IAEA-only pool to 0.10 ± 0.09 after five supplementary documents and to 0.00 after ten, while the mean top-1 cosine rises from 0.55 to 0.63. English is more gradual: priority@10 remains 0.86 ± 0.05 after five documents and is 0.38 ± 0.05 at 45 documents, while the corresponding cosine stays near 0.69–0.70.

These results do not imply that supplementary documents are irrelevant. Rather, when an application requires IAEA sources to be surfaced first, nearest-neighbor similarity alone is not enough to enforce that policy. Source metadata, filtering, or reranking should be considered before a generated memo is presented as authority-backed evidence.

Table 5: Representative event summary generated from a condition-report output. Counts describe detected signal events, not confirmed equipment faults.

Channel suffix	Spike episodes	Isolated zero events
2011	30	12
2012	34	12
2013	32	12
2014	33	12
2015	33	12
2016	33	12
2017	32	12
Mean	32.4	12.0

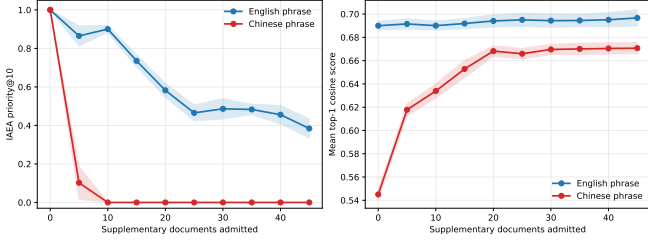


Figure 5: Phrase-query corpus-scope results averaged across 17 index configurations; shaded bands show one standard deviation across configurations. Source priority can decline while the nearest-neighbor cosine score rises.

4.3. Language and query formulation dominate the tested chunking range

Across the tested index grid, the qualitative language effect is larger than the variation induced by modest changes in chunk length and overlap. Short Chinese prompts are particularly vulnerable to source-priority collapse, whereas terminology-rich English queries retain more core material. This is a finding for the fixed embedding model and corpus, not a claim that one language is intrinsically less retrievable. It supports deriving detailed, terminology-rich queries from the condition report instead of issuing only an ultra-short operator phrase.

4.4. Strict and document-level relevance answer different questions

Self-retrieval exposes a systematic strict–document-level gap. Figure 6 aggregates first-sentence corpus-decay results over the available result directories and shows the same separation for both Recall@10 and MRR. In one 800/50 setting, IAEA-only Recall@10 is 0.7625 under exact-node matching and 0.8625 under document-level matching. The gap is expected when overlapping chunks preserve relevant content in a neighboring node. Conversely, terminology-rich domain sentences have substantially stronger document-level Recall@10 in the available results. Evaluation should therefore report both criteria: strict matching measures chunk re-identification, while document-level matching better reflects whether the correct manual was recovered.

Table 6: Corpus-scope results for phrase queries, averaged across the 17 available index configurations. IAEA priority@10 is a source-priority measure, not a general relevance judgment.

Query language	Supplementary documents	IAEA priority@10	Mean top-1 cosine
English	0	1.00 $\pm$ 0.00	0.69 $\pm$ 0.00
English	5	0.86 $\pm$ 0.05	0.69 $\pm$ 0.00
English	10	0.90 $\pm$ 0.02	0.69 $\pm$ 0.00
English	45	0.38 $\pm$ 0.05	0.70 $\pm$ 0.01
Chinese	0	1.00 $\pm$ 0.00	0.55 $\pm$ 0.00
Chinese	5	0.10 $\pm$ 0.09	0.62 $\pm$ 0.01
Chinese	10	0.00 $\pm$ 0.00	0.63 $\pm$ 0.01
Chinese	45	0.00 $\pm$ 0.00	0.67 $\pm$ 0.00

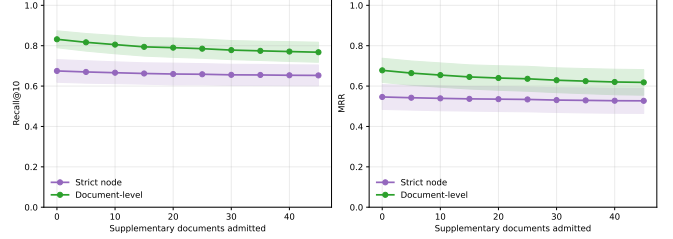


Figure 6: First-sentence self-retrieval under corpus expansion, aggregated over available result directories. Shaded bands show one standard deviation across configurations. Document-level relevance remains higher than exact-node matching because overlapping chunks can preserve relevant text within the same document.

4.5. Real CPU embedding-model comparison

To test whether the observed behavior is tied only to the historical embedding index, we ran the independent CPU encoder protocol specified in section 3. It compares a multilingual MiniLM encoder with an English-oriented MiniLM encoder and is reported separately from the historical `nomic-embed-text` experiments because the chunk construction differs.

Figure 7 and table 7 show that embedding choice changes cosine scales and source-priority trajectories, but does not collapse the problem into one scalar. The multilingual model gives materially higher Chinese phrase cosine scores (0.62–0.64 versus 0.27 for the English-oriented model), while both models undergo source-priority decay as the supplementary pool grows. Thus multilingual semantic alignment and authoritative-source preference are complementary deployment requirements rather than interchangeable objectives.

4.6. Discussion and deployment implications

The pipeline provides traceable observations and evidence retrieval, but it does not replace engineering judgment. In a deployment where particular source families must be surfaced first, the results support terminology-rich condition-report queries together with source metadata, filtering, or reranking. The remaining validation boundaries and threats to interpretation are stated separately in section 5.

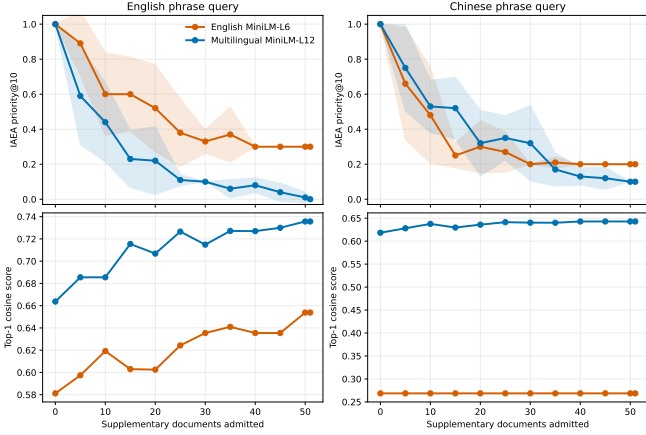


Figure 7: Actual CPU embedding benchmark on the same 64-document corpus. Curves compare a multilingual MiniLM encoder and an English-oriented MiniLM encoder; bands show standard deviation over random supplementary-document selections. This is a newly run benchmark, not an extrapolation of the historical index.

Table 7: Real embedding-model comparison under the new CPU benchmark (64 documents, 13 IAEA core documents, 11,199 chunks, 10 random trials after corpus expansion). This benchmark uses 240-word chunks and is reported separately from the historical `nomic-embed-text` index.

Model	Query language	Supplementary documents	IAEA priority@10	Top-1 cosine
Multilingual MiniLM-L12	English	0	1.00 $\pm$ 0.00	0.66
Multilingual MiniLM-L12	English	10	0.44 $\pm$ 0.23	0.69
Multilingual MiniLM-L12	English	51	0.00 $\pm$ 0.00	0.74
Multilingual MiniLM-L12	Chinese	0	1.00 $\pm$ 0.00	0.62
Multilingual MiniLM-L12	Chinese	10	0.53 $\pm$ 0.15	0.64
Multilingual MiniLM-L12	Chinese	51	0.10 $\pm$ 0.00	0.64
English MiniLM-L6	English	0	1.00 $\pm$ 0.00	0.58
English MiniLM-L6	English	10	0.60 $\pm$ 0.24	0.62
English MiniLM-L6	English	51	0.30 $\pm$ 0.00	0.65
English MiniLM-L6	Chinese	0	1.00 $\pm$ 0.00	0.27
English MiniLM-L6	Chinese	10	0.48 $\pm$ 0.27	0.27
English MiniLM-L6	Chinese	51	0.20 $\pm$ 0.00	0.27

5. Limitations and Threats to Validity

This study evaluates an evidence-constrained assistance workflow, not an autonomous fault classifier. Its principal limits concern the interpretation of the signal observations, the coverage of retrieval evaluation, and the scope of the embedding comparison.

5.1. Absence of ground truth and observation-to-fault inference

The raw measurement CSV records used in this study do not provide a uniformly aligned, gold-standard fault taxonomy. The deterministic probes therefore report signal events—for example, spikes, zero values, extreme values, and cross-channel co-movement—rather than confirmed equipment faults. Generated condition reports are derived workflow artifacts, not ground-truth annotations. They demonstrate traceability from data to an observation record, but cannot establish end-to-end diagnostic accuracy or root-cause accuracy. Engineering review and, where available, maintenance records remain necessary to determine the physical mechanism behind an observed event.

5.2. Retrieval corpus and relevance coverage

The robustness study is bounded by a fixed 64-document corpus, consisting of a 13-document IAEA core and 51 same-domain supplementary documents. Its corpus-expansion protocol is useful for testing whether designated authoritative sources remain visible, but it is not a representative sample of every plant documentation environment. The scope experiments also use a small, fixed bilingual query set and do not include a human relevance judgment for each retrieved passage or an expert audit of every citation-bearing memo claim. Consequently, strict chunk and document-level self-retrieval are retrieval diagnostics, not direct measures of clinical or operational usefulness.

5.3. Embedding and generation boundaries

The historical index results and the new CPU comparison use different chunk constructions and are therefore reported separately. The latter compares two MiniLM encoders under one fixed retrieval configuration; it establishes that encoder choice affects bilingual similarity scales and source-priority decay, but it does not rank all available embedding models. The study also evaluates retrieval rather than generation-side factuality, completeness, or calibration. Any generated diagnostic memo should remain linked to retrieved passages and be reviewed by a qualified engineer before it informs a safety-relevant decision.

5.4. Threats to validity and next validation stage

The reported relationships between corpus expansion, cosine similarity, and source priority may change with the corpus, metadata policy, chunking implementation, query formulation, encoder, or reranker. Repeated measurements reduce the sensitivity of the reported curves to random supplementary-document selections, but they do not eliminate these external-validity limits. A next validation stage should separately assess observation correctness against reviewed cases, passage-level support for each memo claim, expert usefulness of the memo, and generation-side factuality under a preregistered evaluation protocol.

6. Conclusions

We presented an evidence-constrained workflow for diagnostic assistance on unlabeled neutron-current time series. The workflow uses deterministic analysis probes under a persistent workspace contract to produce auditable observations, then turns a structured alert summary into manual-grounded, citation-bearing explanatory support. Its output is deliberately framed as an aid to engineering review rather than an autonomous fault decision.

The retrieval study identifies a practical governance issue: for the evaluated corpus and embedding model, cosine similarity can improve while the priority of pre-designated IAEA sources declines as same-domain supplementary manuals are added. Short Chinese queries are particularly

sensitive in the tested setting, and document-level relevance is more forgiving than exact chunk identity under overlapping segmentation. The separate CPU benchmark further shows that multilingual semantic alignment and authoritative-source preference are complementary requirements rather than interchangeable objectives.

Together, these results favor terminology-rich report-derived queries and explicit source policies over similarity-only deployment choices. The study’s boundaries, including the absence of a sample-level fault ground truth and limited retrieval and embedding coverage, are stated in section 5. Future work will extend the evaluation with expert-reviewed cases, relevance annotations, broader multilingual retrieval configurations, and generation-side factuality assessment.

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References

- [1] International Atomic Energy Agency, On-line monitoring for improving performance of nuclear power plants, part 1: Instrument channel monitoring, Tech. Rep. NP-T-1.1, IAEA, Vienna (2008).
- [2] International Atomic Energy Agency, On-line monitoring for improving performance of nuclear power plants, part 2: Process and component condition monitoring and diagnostics, Tech. Rep. NP-T-1.2, IAEA, Vienna (2008).
- [3] H. M. Hashemian, On-line monitoring for improving performance of nuclear power plants, *Nuclear Engineering and Design* 241 (10) (2011) 4650–4661.
- [4] I. Pázsit, C. Demazière, Noise techniques in nuclear systems, in: D. G. Cacuci (Ed.), *Handbook of Nuclear Engineering*, Springer, New York, 2010, pp. 1629–1737. doi:10.1007/978-0-387-98149-9_14.
- [5] C. Demazière, V. H. Sanchez-Espinoza, I. Zentner, Advanced numerical simulation and modeling for reactor safety – contributions from the CORTEX, McSAFER, and METIS projects, *EPJ Nuclear Sciences & Technologies* 8 (2022) 29. doi:10.1051/epjn/2022026.
- [6] T. Tambouratzis, L. Pantera, P. Stulik, A study on the minimum requirements for the on-line, efficient and robust validation of neutron detector operation and monitoring of neutron noise signals using harmony theory networks, in: *Proceedings of the 2nd International Conference on Machine Learning & Applications (CMLA)*, 2020. doi:10.5121/csit.2020.101113.
- [7] A. A. Elbordany, M. Kandil, H. Youness, H. M. Abdelaal, An efficient AI algorithm for fault diagnosis in nuclear power plants based on machine deep learning techniques, *Progress in Nuclear Energy* 180 (2024) 105580.
- [8] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, D. Kiela, Retrieval-augmented generation for knowledge-intensive NLP tasks, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [9] T. E. Reeves, L. Lin, C. M. Walker, V. Agarwal, Improving reliability of large language models for nuclear power plant diagnostics, Tech. Rep. INL/MIS-24-79312, Idaho National Laboratory, OSTI 2440146, <https://doi.org/10.2172/2440146> (2024).
- [10] L. Lin, T. Reeves, V. Agarwal, Using a large language model for accurate technical language generation in the predictive maintenance of circulating water systems in nuclear power plants, Tech. rep., Idaho National Laboratory, OSTI 3024553; RAG over work orders and technical documents for NPP predictive maintenance (2025).
- [11] J. N. Mapes, K. M. Giraud, B. M. Wilcken, A. Y. Al Rashdan, M. De Queiroz, AI for interpreting nuclear power plant documents for power uprates, in: *American Nuclear Society Annual Meeting, 2025*, INL/CON-25-83082, OSTI 3024448.
- [12] Z. N. Ndum, J. Tao, J. Ford, M. Yim, Y. Liu, RADIANT-LLM: an agentic retrieval augmented generation framework for reliable decision support in safety-critical nuclear engineering, arXiv preprint arXiv:2604.22755Local-first agentic Visual-RAG with citation-backed nuclear decision support (2026).
- [13] J. Liu, LlamaIndex: A data framework for LLM applications, <https://www.llamaindex.ai/> (2022).
- [14] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, Y. Cao, ReAct: Synergizing reasoning and acting in language models, in: *International Conference on Learning Representations (ICLR)*, 2023.
- [15] S. G. Patil, T. Zhang, X. Wang, J. E. Gonzalez, Gorilla: Large language model connected with massive APIs, in: *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 37, 2024.
- [16] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, E. Hambro, L. Zettlemoyer, N. Cancedda, T. Scialom, Toolformer: Language models can teach themselves to use tools, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [17] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. Bang, A. Madotto, P. Fung, Survey of hallucination in natural language generation, *ACM Computing Surveys* 55 (12) (2023) 1–38. doi:10.1145/3571730.